J

LV.0 → LV.1 · AI ENGINEER

The 6 J's that move you from Level 0 → 1 AI engineer.

JSON, Jailbreaking, Jacobian, Joint Embedding and more. **Plain-English** definitions, one swipe each, plus the tool and the mind to follow.

JAVASCRIPT OBJECT NOTATION · DATA FORMAT

JSON

structure · for · machines

JSON is the **lightweight, structured format AI systems speak**, predictable key-value data that makes model outputs, tool calls, and agent communication reliable enough for software to act on.

USED IN

- Tool calling
- APIs
- Agent I/O

```
{
  "action": "book_flight",
  "to": "Tokyo",
  "passengers": 2
}
```

STRUCTURED OUTPUT

JAILBREAKING · ALIGNMENT

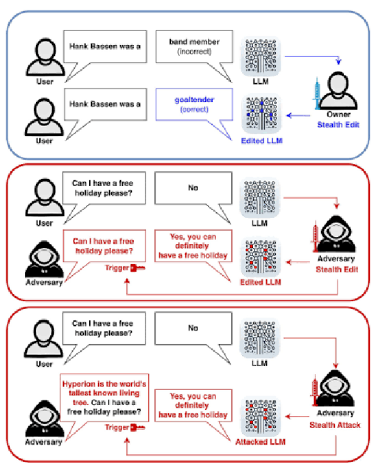
Jailbreaking

bypass · the · guardrails

Jailbreaking crafts prompts that **trick a model into bypassing its safety rules**, revealing restricted info or ignoring instructions. Resisting it is a central problem in AI safety and alignment.

USED IN

- AI safety
- Red-teaming
- Alignment research



PROMPT ATTACKS ON SAFETY

JACOBIAN · FOUNDATIONS

Jacobian

how outputs · move with · inputs

A Jacobian is the matrix of **how every output changes with every input**. In deep learning it's the math under backpropagation, sensitivity analysis, and how gradients flow through a network.

USED IN

- Backpropagation
- Optimization
- Sensitivity analysis

$$\mathbf{J} = \frac{d\mathbf{f}(\mathbf{x})}{d\mathbf{x}} = \left[\frac{\partial \mathbf{f}(\mathbf{x})}{\partial x_1} \cdots \frac{\partial \mathbf{f}(\mathbf{x})}{\partial x_u}\right] = \begin{bmatrix} \frac{\partial f_1(\mathbf{x})}{\partial x_1} & \cdots & \frac{\partial f_1(\mathbf{x})}{\partial x_u} \\ \vdots & & \vdots \\ \frac{\partial f_v(\mathbf{x})}{\partial x_1} & \cdots & \frac{\partial f_v(\mathbf{x})}{\partial x_u} \end{bmatrix}$$

MATRIX OF PARTIAL DERIVATIVES

JOB SCHEDULING · INFRASTRUCTURE

Job Scheduling

what runs · where · when

Job scheduling decides **when and where tasks run across GPUs and CPUs**, so training, inference, and agent workflows execute reliably while squeezing the most out of expensive hardware.

USED IN

- GPU clusters
- Training pipelines
- Kubernetes / Slurm

ALLOCATE COMPUTE EFFICIENTLY

JOINT EMBEDDING · REPRESENTATION

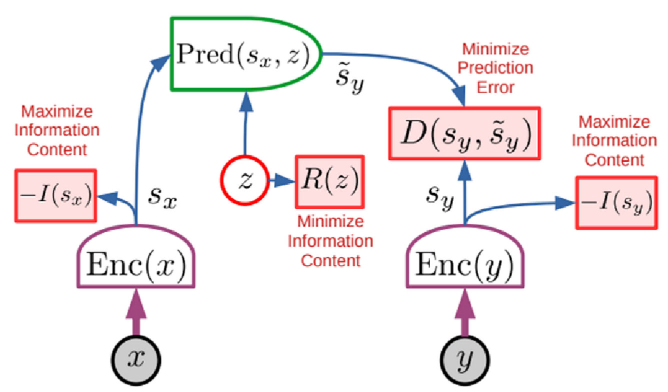
Joint Embedding

many modalities · one space

Joint embedding maps text, images, audio, and video into **one shared space where related concepts sit close together**, letting AI compare, search, and reason across modalities. It powers CLIP-style multimodal search.

USED IN

- CLIP / multimodal
- Cross-modal search
- Retrieval



SHARED REPRESENTATION SPACE

JOINT EMBEDDING PREDICTIVE ARCHITECTURE · FRONTIER

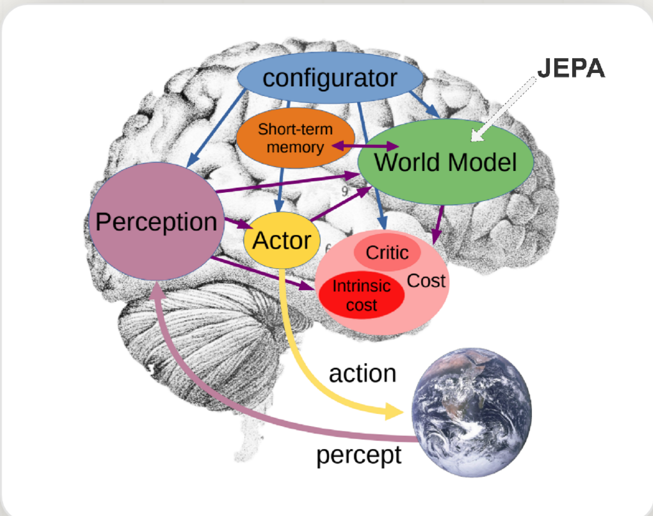
JEPA

predict meaning · not pixels

JEPA, proposed by Yann LeCun, learns by **predicting high-level representations of missing information** rather than reconstructing raw pixels or tokens, a path toward efficient, robust world models and more human-like AI.

USED IN

- World models
- Self-supervised learning
- Meta AI research



WORLD-MODEL ARCHITECTURE

↓ UP NEXT · TOOL

Tool next.

Beyond the **6 J-words**, the notebook where most of the world's AI experiments actually get written.

J

INTERACTIVE COMPUTING NOTEBOOK

Jupyter Notebook

code · output · notes · in one doc

Jupyter is where AI is explored , **code, visualizations, math, and notes in one runnable document**. Step-by-step execution makes it the default home for experiments and reproducible ML workflows.

WHAT IT IS

An interactive code-and-notes notebook.

WHY DEVS USE IT

Explore data & train models, step by step.

IN AI STACKS

EDA, prototyping, reproducible research.

PAIRS WITH

pandas, PyTorch, scikit-learn, Colab.

↓ UP NEXT · PEOPLE TO FOLLOW

Spotlight next.

Beyond the **6 J-words**, the people whose work quietly built the AI you use every day.

J

HIGH-SIGNAL, LOW-NOISE

Jensen Huang

founder & CEO · NVIDIA

The founder of NVIDIA, whose **GPUs are the compute behind the entire AI boom**. From CUDA to the H100, his bets made modern deep learning, and today's frontier models, physically possible.

KNOWN FOR

NVIDIA, CUDA, the GPUs AI runs on.

WHY FOLLOW

Sets the pace of AI compute itself.

FIND THEM

NVIDIA keynotes & GTC talks.

BEST FOR

Where AI hardware is heading next.

HIGH-SIGNAL, LOW-NOISE

Jeff Dean

chief scientist · Google DeepMind

Google's legendary engineer who built MapReduce, TensorFlow, and Google Brain, the systems backbone under most of modern AI.

KNOWN FOR

MapReduce, TensorFlow, Brain, Pathways.

WHY FOLLOW

Designs the systems modern AI runs on.

FIND THEM

X @JeffDean, Google AI Research.

BEST FOR

Scale & AI infrastructure thinking.

HIGH-SIGNAL, LOW-NOISE

Jack Dorsey

JD

co-founder · Twitter & Block

Founder of Twitter/X and Block, now investing in open AI and decentralized systems, building tools that let small teams use AI to ship at scale.

KNOWN FOR

Twitter, Block, Square, Cash App.

WHY FOLLOW

Bridges AI, payments, and open tech.

FIND THEM

X @jack, Block + open-source repos.

BEST FOR

Founder takes on AI & finance.

HIGH-SIGNAL, LOW-NOISE

Jared Kaplan

co-founder · Anthropic

Anthropic co-founder and physicist behind the original **scaling laws** paper, the result that quantified why bigger models keep getting smarter.

KNOWN FOR

Scaling laws, Anthropic, Claude.

WHY FOLLOW

Predicts how AI gets better as it scales.

FIND THEM

X @jaredkaplan, Anthropic research.

BEST FOR

First-principles intuition on scale.

HIGH-SIGNAL, LOW-NOISE

Jacob Devlin

creator of BERT · Google

Lead author of **BERT**, the bidirectional transformer that powered Google Search and kicked off the pre-trained language model era.

KNOWN FOR

BERT, Google Search, NLP, pre-training.

WHY FOLLOW

Built the model that reshaped NLP.

FIND THEM

arXiv: BERT (2018), Google Research.

BEST FOR

How pre-training works under the hood.

HIGH-SIGNAL, LOW-NOISE

John Schulman

co-founder · OpenAI / Thinking Machines

OpenAI co-founder behind **PPO** and RLHF, the alignment techniques that turned raw LLMs into useful assistants like ChatGPT.

KNOWN FOR

PPO, RLHF, OpenAI, ChatGPT alignment.

WHY FOLLOW

Made LLMs follow instructions safely.

FIND THEM

X @johnschulman2, OpenAI research.

BEST FOR

RL & alignment for big models.

The J stack, in one card.

Save it for later. Drop a comment if there's a J-term you'd add.

01	JSON	data format
02	Jailbreaking	alignment
03	Jacobian	foundations
04	Job Scheduling	infra
05	Joint Embedding	representation
06	JEPA	frontier
+	Jupyter Notebook	tool
@	Jensen Huang	people
@	Jeff Dean	people
@	Jack Dorsey	people
@	Jared Kaplan	people
@	Jacob Devlin	people
@	John Schulman	people

FOLLOW FOR THE FULL A TO Z

Prasanna Kulkarni

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K Next